

Assignment - 9

Section - A

1 Define a line integral of a complex fn.
let $f(z)$ be a complex fn. continuous on a curve C in the complex plane

if the curve C is represented by
 $z = z(t), a \leq t \leq b$

then the line integral of $f(z)$ along C is defined as

$$\int_C f(z) dz = \int_a^b f[z(t)] z'(t) dt$$

2 What is meant by a contour in the complex plane?
A contour in the complex plane is directed piecewise smooth curve made up of a finite number of smooth arcs joined together.

It represents a path along which a complex function is integrated.

3 State Cauchy-Goursat theorem

If a function $f(z)$ is analytic everywhere inside and on a simple closed contour C , then

$$\oint_C f(z) dz = 0.$$

4 Define an isolated singularity of a complex function.
A point $z = a$ is called an isolated singularity of a complex fn $f(z)$ if:

→ $f(z)$ is not analytic at $z = a$, but

→ $f(z)$ is analytic in some neighbourhood around $z = a$ except at the point itself.

5/ What is zero of an analytic fn.

A point $z=a$ is called a zero of an analytic fn. $f(z)$ if $f(a)=0$

→ i.e. the fn. value becomes zero at the point.

Section-B

1/ Evaluate the line integral

$$\int_C (z^2+1) dz$$

where C is straight line joining $z=0$ to $z=1+i$
since z^2+1 is analytic everywhere, we as

$$\oint f(z) dz = f(z) \Big|_{z_1}^{z_2}$$

where $f'(z) = f(z)$

Integrating, $f(z) = \frac{z^3}{3} + z$

$$\int (z^2+1) dz = \left[\frac{z^3}{3} + z \right]_0^{1+i}$$

$$\frac{(1+i)^3}{3} + (1+i) = \frac{1+5i}{3} \quad \underline{\underline{\text{Ans}}}$$

2/ Calculate $\oint \frac{z^2}{z-1} dz$ $C: |z-1|=2$

The contour $|z-1|=2$ is a circle centered @ $z=1$ with radius 2.

The integrand has singularity at
 $z=1$

using Cauchy Integral Formula :

$$\oint \frac{f(z)}{z-a} dz = 2\pi i f(a)$$

$$f(z) = z^2, a = i \Rightarrow f(a) = (i)^2 = -1$$

$$\oint \frac{f(z)}{z-a} dz = 2\pi i (-1) \\ = \underline{2\pi i}$$

3 Find the zeros of the fn.

$$f(z) = z^2 - 4$$

and state their order.

$$f(z) = 0$$

$$z^2 - 4 = 0$$

$$(z-2)(z+2) = 0$$

Hence the zeros are :

$$z = 2, z = -2$$

Each factor occurs only once, therefore each zeros of order 1.

Part - C

1 Evaluate the contour integral

$$\oint \frac{e^z}{z-2} dz$$

where C is the circle $|z-2| = 1$, using Cauchy - Integral formula.

$$\oint \frac{f(z)}{z-a} dz = 2\pi i f(a)$$

$$f(z) = e^z, \quad a = 2$$

$$\therefore \oint \frac{e^z}{z-2} dz = 2\pi i e^2$$

$$= \boxed{2\pi i e^2} \quad \text{Ans}$$

2. Let $f(z) = 3+2i$, determine whether $f(z)$ satisfies the conditions of Liouville's Theorem and state the nature of the function.

Liouville's Theorem

Statement:

If a function $f(z)$ is:
 1. analytic everywhere in the complex plane and
 2. Bounded throughout the complex plane
 then $f(z)$ must be a constant function.

Given fn: $f(z) = 3+2i$

→ no variable

→ Constant Complex fn.

Check analyticity

$$\frac{d}{dz} (3+2i) = 0$$

Hence $f(z)$ is analytic for all $z \in \mathbb{C}$

therefore $f(z)$ entire

check Boundedness

The modulus of the fn. is

$$|f(z)| = |3+2i|$$

$$|a+ib| = \sqrt{a^2+b^2}$$

$$\text{So, } |3+2i| = \sqrt{3^2+2^2} = \sqrt{13}$$

$$|f(z)| = \sqrt{13}$$

which is a fixed finite number
Hence the fn is bounded.

Apply Liouville's Theorem
Since $f(z)$ is:

→ entire and bounded

Liouville's theorem applies

∴ ~~$f(z)$~~ must be constant fn.

∴ $f(z) = 3+2i$, satisfies Liouville's theorem

3 Define the maximum modulus theorem and
determine whether the function $f(z) = z^2$ can
attain its maximum value inside the region $|z| < 1$
maximum modulus theorem

If $f(z)$ is analytic and non-constant in a
domain D , then the modulus

$$|f(z)|$$

cannot attain its maximum value at any
interior point of D .

The maximum value occurs only on the boundary of the region

given fn: $f(z) = z^2$

- polynomial fn.
- analytic everywhere in complex plane
- It's non-constant
- ∴ maximum modulus theorem is applicable

$$|f(z)| = |z^2|$$

$$|z^2| = |z|^2$$

hence $|f(z)| = |z|^2$

use the condition $|z| < 1$

inside the region, $|z| < 1$

Squaring both sides $|z|^2 < 1$

$$\therefore |f(z)| < 1$$

Check boundary on the boundary $|z| = 1$

$$\therefore |f(z)| = |z|^2 = 1$$

Thus the maximum value is attained on the boundary circle.